

11/11/11-06-11

Q-24

Write down the expression for M-I of uniform hemisphere of radius a about its axis of symmetry, where $dV = dr(r d\theta)(r \sin\theta d\phi) = r^2 \sin\theta dr d\theta d\phi$.

Note: only write expression.

M-I

Ans:-

$$I_{zz} = \rho \int_0^a \int_0^{\pi/2} \int_0^{2\pi} r^4 \sin^3\theta dr d\theta d\phi$$

$$= \frac{2}{5} \rho a^2$$

Ans-

Q-25

Use the

M-I

line of

to axis

Sol:-

by p.d

since

is

by ①

I

Example: If

$$\sum a_n = \sum \frac{n!}{\alpha(\alpha+1)(\alpha+2)\cdots(\alpha+n-1)}, \quad \alpha > 0,$$

then

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{n+1}{\alpha+n} = 1,$$

so the ratio test is inconclusive.



2.14. Cauchy's Root Test

However,

$$\begin{aligned} \lim_{n \rightarrow \infty} n \left(\frac{a_{n+1}}{a_n} - 1 \right) &= \lim_{n \rightarrow \infty} n \left(\frac{n+1}{\alpha+n} - 1 \right) \\ &= \lim_{n \rightarrow \infty} \frac{n(1-\alpha)}{\alpha+n} = 1-\alpha, \end{aligned}$$

so Raabe's test implies that $\sum a_n < \infty$ if $\alpha > 2$ and $\sum a_n = \infty$ if $0 < \alpha < 2$.

Theorem: If f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) , then

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

for some c in (a, b) .



4.7. Generalized Mean Value Theorem

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Proof: The function

$$h(x) = [b - a]f(x) - [f(b) - f(a)]x$$

is continuous on $[a, b]$ and differentiable on (a, b) .

Furthermore

$$h(a) = h(b) = bf(a) - f(b)a.$$

Therefore, Rolle's theorem implies that $h'(c) = 0$ for some c in (a, b) .

$$\text{Since } h'(c) = [b - a]f'(c) - [f(b) - f(a)].$$

Example: Consider the series

$$\sum \frac{\sin n\theta}{n^p},$$

where θ is arbitrary and $p > 1$.

Solution: Since

$$\left| \frac{\sin n\theta}{n^p} \right| \leq \frac{1}{n^p}$$

and $\sum 1/n^p < \infty$ if $p > 1$, the comparison test implies that

$$\sum \left| \frac{\sin n\theta}{n^p} \right| < \infty, \quad p > 1.$$

Therefore, the given series converges absolutely if $p > 1$.

Example: If $0 < p < 1$, then the series

$$\sum \frac{(-1)^n}{n^p}$$

does not converge absolutely, since

$$\sum \left| \frac{(-1)^n}{n^p} \right| = \sum \frac{1}{n^p} = \infty.$$

However, the series converges, by the alternating series test, which we prove b

Remark: Any test for convergence of a series with nonnegative terms can be used to test an arbitrary series $\sum a_n$ for absolute convergence by applying it to $\sum$

Example: Determine the series

$$\sum a_n = \sum (-1)^n \frac{n!}{\alpha(\alpha+1) \cdots (\alpha+n-1)}, \quad \alpha > 0,$$

for absolutely convergent or not?

2.17. Dirichlet's Test for Series

We apply Raabe's test to

$$\sum a_n = \sum \frac{n!}{\alpha(\alpha+1) \cdots (\alpha+n-1)}.$$

From previous example, $\sum |a_n| < \infty$ if $\alpha > 2$ and $\sum |a_n| = \infty$ if $\alpha < 2$. Therefore $\sum a_n$ converges absolutely if $\alpha > 2$, but not if $\alpha < 2$.

Notice that this does not imply that $\sum a_n$ diverges if $\alpha < 2$.

Theorem: If f is integrable on $[a, b]$ and c is a constant, then cf is integrable on $[a, b]$ and

$$\int_a^b cf(x) dx = c \int_a^b f(x) dx.$$

Proof: See Lecture.

Now from definition $S(P)$ is the supremum of the set of Riemann sums of f over P .

Example: Let

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is irrational,} \\ 1 & \text{if } x \text{ is rational,} \end{cases}$$

Let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$. Since every interval contains both rational and irrational numbers

$$m_j = 0 \quad \text{and} \quad M_j = 1, \quad 1 \leq j \leq n.$$

Hence,

$$\begin{aligned} S(P) &= \sum_{j=1}^n 1 \cdot (x_j - x_{j-1}) = b - a \\ s(P) &= \sum_{j=1}^n 0 \cdot (x_j - x_{j-1}) = 0. \end{aligned}$$

Since all upper sums equal $b - a$ and all lower sums equal 0, from definition we have

$$\overline{\int_a^b} f(x) dx = b - a \quad \text{and} \quad \underline{\int_a^b} f(x) dx = 0.$$

Theorem: If f is differentiable at x_0 , then f is continuous at x_0 .

Proof:

$$f(x) = f(x_0) + [f'(x_0) + E(x)](x - x_0),$$

where E is defined on a neighborhood of x_0 and

$$\lim_{x \rightarrow x_0} E(x) = E(x_0) = 0.$$

where

$$E(x) = \begin{cases} \frac{f(x) - f(x_0)}{x - x_0} - f'(x_0), & x \in D_f \text{ and } x \neq x_0, \\ 0, & x = x_0. \end{cases}$$

Example: Show that the function $g(x) = x^2$ is uniformly continuous on $[-r, r]$ for any finite r .

Solution: To see this, note that

$$|g(x) - g(x')| = |x^2 - (x')^2| = |x - x'| |x + x'| \leq 2r|x - x'|,$$

so

$$|g(x) - g(x')| < \varepsilon \quad \text{if} \quad |x - x'| < \delta = \frac{\varepsilon}{2r} \quad \text{and} \quad -r \leq x, x' \leq r.$$